

Sam McFarland

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EDUCATION

Vanderbilt University

B.S. in Computer Science, Minor in Data Science (GPA: 4.00)

Nashville, TN

August 2025 – May 2028

EXPERIENCE

Vanderbilt Dry Dock Venture Studio — 1st place, \$45,000

Nashville, TN

Founding Software Engineer

May 2026 – June 2026

- Created a full-stack wireless spectrum trading platform (FastAPI, React/Next.js, TypeScript, Supabase/Postgres) in 1 week.
- Implemented an exact VCG combinatorial auction (NP-hard 2^{13} bitmask-DP WDP, 10 ms/round) to ensure truthful bidding.
- Built 15 Python ETL pipelines loading GB-scale data (FCC, Census, Ookla, NYC Taxi) into H3-indexed PostgreSQL tables.
- Designed deterministic pricing engine calibrated to FCC auction data, blended with live trades, covered by 143 unit tests.

TECHNICAL PROJECTS

Scramble — Daily Word Game

Nashville, TN

Founder & Software Engineer

May 2025 – Present

- Built and submitted SwiftUI iOS app to App Store, serving 50 TestFlight users with Firebase and Cloud Functions backend.
- Replaced an $O(n \cdot n!)$ permutation word-solver with a trie-backed pruned DFS, cutting puzzle generation time by 12.5x.
- Added peppered HMAC-SHA256 contact matching in Node.js, enabling friend discovery without storing phone numbers.
- Wrote 200+ lines of Firestore rules with field whitelists and server-time checks, closing score-forgery and TOCTOU races.

Ancori

Nashville, TN

Founding Software Engineer

April 2026 – June 2026

- Built and deployed Ancori, a Next.js/TypeScript web-app that turns any website's design into .md files for AI coding tools.
- Architected component system of 16 layout templates rendering 14 design systems from JSON, cutting setup to one file.
- Wrote dependency-free markdown renderer parsing tables, code blocks, and nested lists, escaping input to stop injection.
- Shipped Google OAuth with gated previews, 5 export targets, and server-rendered routes indexed for SEO on a live site.

OneTap — VandyHacks XII

Nashville, TN

Software Engineer

March 2026 – April 2026

- Won Best Startup and Best Use of MongoDB Atlas at a 100+ participant 24-hour hackathon sponsored by Capital One.
- Built a Chrome Manifest V3 extension that detects checkout pages and auto-fills the best Capital One card in one tap.
- Wrote a 139-selector autofill engine using native setter overrides to fill React-controlled inputs inside checkout iframes.
- Designed 6 MongoDB Atlas collections with 12 compound indexes backing a 20-endpoint Node/Express REST API.

WolfTrade

Nashville, TN

Software Engineer

March 2026 – April 2026

- Built a stock charting web app that teaches beginners to read charts, unlocking 15 features across 10 completed lessons.
- Wrote an SVG overlay engine rendering 5 drawing tools, syncing shapes to price-time coords across pan/zoom repaints.
- Architected a 10-lesson, 32-slide curriculum as a typed schema rendering 13 animated SVG diagrams from one data file.
- Proxied Alpha Vantage behind Next.js route with regex symbol sanitizing, 5-min ISR caching, and mapped 429/404 errors.

LEADERSHIP

Vanderbilt Innovation and Entrepreneurship Society (VINES)

Nashville, TN

Vice President of Professional Development

January 2026 – Present

- Interviewed 100+ applicants through technical and behavioral assessments and led a mentor program for new members.
- Taught entrepreneurship sessions for 20+ students covering venture capital, customer discovery, and business models.

FRead Inc.

Newton, MA

Founder & President

June 2024 – August 2025

- Founded a 20-member nonprofit that provides free children's books in 5 health offices, including Mass General Hospital.

Skills

Languages: Python, Java, Swift, C++ **Technologies:** React, Next.js, FastAPI, Node.js, Firebase, Git, Postgres, Claude Code